

Application Tracking System (ATS) :-

Tech Stack (Java Backend)

- * Framework: Spring Boot
- * Database: MySQL
- * ORM: Hibernate (Spring Data JPA)
- * Authentication: Spring Security + JWT / OAuth2 (LinkedIn, Google)
- * API style: REST
- * File Upload: Spring Multipart Support-

Week 01: Project- Setup and Core Foundation :-

* Goals :-

- ① Set up environment, tools and base Spring Boot project.
- ② Design database schemas.
- ③ Implement user authentication (JWT)

↳ Task :

- ① Development Setup :- Install java, Spring Boot, Maven / Gradle, MySQL
↳ Use Postman or Swagger for API Testing
↳ Set up GitHub repo.

② Spring Boot Initialization :-

- ↳ Use Spring initializer with: Spring web, Spring security, Spring Data JPA, MySQL Driver, Lombok

③ Database Design :- FRD entities with:

user, token, jobs, Role, Permission, Role-Permission, user-user, feedback, job application, notification, meeting, Resume.

④ Authentication & Authorization :-

- ① JWT-based login / register
- ② RBAC: Admin, HR, interview, candidate
- ③ Use Spring security with custom filters.

Week 02: Job Management and Candidate Portal.

* Goals :- ↳ Implement CRUD operations on job posts.

④ Authentication & Authorization :-

- ① JWT-based login / register
- ② RBAC: Admin, HR, interview, candidate
- ③ Use Spring security with custom filter.

Week 02: Job Management and Candidate Portal.

* Goals :- $\rightarrow$ Implement CRUD operations for job posts.

$\rightarrow$ Enable candidate registration, profile update and job application.

Tasks :- ① Job Management :-

- $\rightarrow$ Create job entities, repository, service, and controller.
- $\rightarrow$ Endpoint for HR/Admin: create, update, delete and view jobs.
- $\rightarrow$ Add filtering by department, location, tags.

② Candidate Portals :-

- $\rightarrow$ Candidate registration and profile update.
- $\rightarrow$ Apply to jobs: upload resume (store file path or blob).

↳ store application linked to user & jobs.

③ Input validation & Exception Handling:-

↳ use @Valid, @NotBlank and a global exception handler.

Week 03: AI Resume Scoring and Interview Management

* Goals:- ① Integrated with python-based AI scoring microservices.

↳ Implement interview scheduling and feedback.

↳ Tasks:- ① Resume Scoring Integration:-

↳ REST API to call python microservices / score-resume.

↳ Send resume-text + job description.

↳ Receive match score and store in DB.

② Store Resume Scores:-

↳ Save matched-keywords, missing-keywords, score in application entity.

③ Interview Scheduling:-

↳ Tasks:- ① Resume Scoring Integration:-

- ↳ REST API to call python microservices / score-resume.
- ↳ Send resume text + job description.
- ↳ Receive match score and store in DB.

② Store Resume Scores:-

- ↳ Save matched keywords, missing keywords, score in application entity.

③ Interview Scheduling:-

- ↳ Schedule interviews
- ↳ Assign interview
- ↳ Record feedback (rating, notes)

④ Email Notification System:-

- ↳ Use JavaMail Sender or integrates SMTP services.

Week 04:- Candidate History, Analyses & Reports:

* Goals:-

- ↳ implement history tracking and basic dashboard data.
- ↳ provide HR and Admin tools to view reports.

* Tasks :-

① Candidate History :-

↳ Timeline view of application, interview, feedback.

② Dashboard API :- Job status overview, Applicant funnel, hiring trends.

③ Reports & Exports :- API to exports CSVs for job application, interview outcome.

④ Role-Based Dashboards :- Admin vs HR vs Interview view

Week 05 :- Final Touches & optional enhancement :-

① Email Reminders ② Exception handler

③ Testing ④ prepare for & Semester
examination.

Module Structure :-

↳ Controller :- ① Job controller

② Candidate controller ③ Auth-
Controller ④ Interview

Deekane.

(b) Role-Based dashboards :- Admin vs HR
vs Interview view

Week 05 :- Final Touches & optional
enhancement :-

- (1) Email Reminders (2) Exception handler
- (3) Testing (4) prepare for & security
deployment.

Modular structure :-

↳ Controller :- (1) Job controller
(2) Candidate controller (3) Auth-
- Controller (4) Interview
Controller.

↳ Service :- (1) Job Services (2) Candidate-
- services (3) Interview services
(4) ResumeService services.

↳ model :- (1) Job (2) Candidate
(3) Application (4) Interview,
(5) Resume controller (6) Role
(7) User.

↳ Repository :- (1) Job Repository,
(2) Candidate Repository

↳ Security :- (1) JwtFilter (2) JwtUnit
(3) Custom user details services.

ls.config :- ① Securityconfig,
② webMvc.config.

↳ dto :- ① ResumeSource DTO, LoginRequest
③ Job DTO ④ Candidate DTO.

Week 01:- Project setup and core Foundation:-

Day 1: Project initialization:-

- ↳ Setup Spring Boot project using Spring Initializer.
- ↳ Chose dependencies.
- ↳ Setup version control (Git repository).
- ↳ create base project structure.

Day 2: Database configuration:-

- ↳ Setup application properties (DB URL, username, password).
- ↳ JPA & Hibernate settings.

Day 03: Create User Entity And Role:-

- ↳ design user entity with basic fields (id, name, email, password, role, createdAt).
- ↳ create Role enum & separate table for user roles. (ADMIN, RECruITER, CANDIDATE)

Day 04: Authentication System - JWT Set

- ① Setup spring security configuration.
- ② create custom UserDetails service.
- ③ configure authentication filter.
- ④ Integrate JWT token generation.

and Login Endpoint

↳ JPA & Hibernate settings

Day 03: Create User Entity and Role

- ↳ design user entity with basic fields.
(id, name, email, password, role, createdAt).
- ↳ create Role enum & separate table for user roles. (ADMIN, RECruITER, CANDIDATE)

Day 04: Authentication System - JWT Set

- ① Setup spring security configuration.
- ② Create custom UserDetails service.
- ③ Configure authentication filter.
- ④ Integrate JWT token generation.

Day 05: Register and Login Endpoint

- ① create /auth/register & /auth/login.
- ② Use DTOs for input/output.
- ③ Implement password encryption using BCrypt.

Day 06: Role-based Authorization

↳ secure endpoint using Spring Security

Day 07: Code Cleanup & Documentation

- ↳ Refactor repeated code (e.g. token utility).
- ↳ Add Swagger/OpenAPI for API docs.

Week 02:

Day 01: Job post entity and Basic Setup.

Day 02: Job post CRUD Endpoint

Day 03: Candidate Registration & Profile Entity.

Day 04: Candidate Profile CRUD.

Day 05: Job Application Entity and Setup

Day 06: Apply to job and Application endpoint —

Day 07: Testing, Swagger, Docs.

Week 03:

Day 01: Resume Scoring - Design & Integration

Day 02: Resume Scoring Engine Implementation

Day 03: Resume parsing

Day 04: Interview entity & setup

Day 05: Interview Scheduling

Week 04:

Day 01: Admin. Panel - Access & Features

Day 02: Application Analytics Backend & Logging

Maggen, Roos.

Week 03:

- Day 01: Resume Scoring - Design & Implement
- Day 02: Resume Scoring Engine Implementation
- Day 03: Resume parsing
- Day 04: Interview entity & setup
- Day 05: Interview Scheduling

Week 04:

- Day 01: Admin. Panel - Access & Features
- Day 02: Application Analytics Backend
- Day 03: User Actions & Logging
- Day 04: Notification System
- Day 05: Frontend or Postman